

# LUKE P. CORCOS

lucorcos@gmail.com \* 1-(410)-430-0185 \* <https://lpcorcos.github.io> \* [LinkedIn](#) \* [Scholar](#) \* *BlockSp*

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Summary: A market-experienced applied mathematician, Ph.D.

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## RESEARCH EXPERIENCE

**Lawrence Berkeley National Laboratory**  
Graduate Student Researcher

2017 - 2023

- Constructed high-fidelity mathematical and numerical models for multi-physics multi-layer coating flow dynamics, with applications to the leveling of multi-layer paint films.
- Developed a tailored dimension-independent high-performance computing (HPC) framework in C++, parallelized via MPI. Simulations were performed on NERSC's *Cori* and *Perlmutter* supercomputers; up to 400,000 elements and 14 million spatial degrees of freedom.
- Collaborated with research scientists from the Lawrence Berkeley National Laboratory (LBNL) and PPG Industries Inc. via the Department of Energy's (DOE) *HPC4Mfg* program. Performed high-resolution parametric studies at industrially-relevant conditions, providing actionable insights on the formation of coating irregularities.

Other coding experience

- Developed and maintain *BlockSp*, a templated C++ library for block-sparse linear algebra. Features include: matrix-matrix and matrix-vector multiply, preconditioned CG and GMRES algorithms, and Arnoldi eigenvalue solvers.
  - Implemented C++ algorithms for capturing rising bubble dynamics, including multigrid methods, level set methods, and projection methods for solving the incompressible Navier-Stokes equations.
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## MARKET EXPERIENCE

**Independent Market Researcher**

2016 - Present

- Actively trade stocks, options, and cryptocurrencies using the CANSLIM method along with self-discovered chart patterns. Operate a blog dedicated to stock trading: <https://lpcorcos.github.io/stocks/>.
  - Achieved a compound annual growth rate (CAGR) of 31% since 2020, generating an average annual alpha of 14.8 w.r.t the S&P 500, obtaining a Sharpe ratio of 0.75. Returned 111% in 2024, outperforming the S&P 500 by 4.75x.
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## PUBLICATIONS

L.P. Corcos, R.I. Saye, J.A. Sethian, *A hybrid finite difference level set-implicit mesh discontinuous Galerkin method for multi-layer coating flows*. J. Comput. Phys. 2024, 507, 112960 (2024).  
A.D. Cimmarusti, Z. Yan, B.D. Patterson, L.P. Corcos, L.A. Orozco, S. Deffner, *Environment-assisted speed-up of the field evolution in cavity quantum electrodynamics*, Physical Review Letters 114 (23), 233602 (2015).

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## EDUCATION

**Ph.D. in Applied Mathematics**

Received 2023

Awarded the Bernard Friedman Memorial Prize in Applied Mathematics.  
Thesis advisor — Professor James A. Sethian  
University of California, Berkeley.

**B.S. in Mathematics and Physics**

Received 2016

High Honors in Physics, Banneker Key Scholar, University Honors.  
University of Maryland, College Park.

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## SKILLS

C++, MPI, MKL, LAPACK, Linux, Make, Bash, Mathematica, MatLab, LaTeX, Office.

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References available upon request.